Reinforcement Learning for Large-Scale Auto-Bidding Challenge

COS 435 / ECE 433 — Course Project

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Problem Setting

Setting: An auto-bidding agent bids on ad slots on behalf of an advertiser given relevant information about the ad opportunities.

Environment: AuctionNet benchmark (NeurIPS 2024 Competition).

Goal: Maximizes the cumulative conversion rate.

Constraints:

- **Total Budget:**
  - Hard Constraint, cannot exceed
- **Cost-per-Action (CPA):**
  - Soft Constraint
  - Penalize reward when exceeded.

Formulation & Baseline

State: 16-feature observation describing ad opportunity.

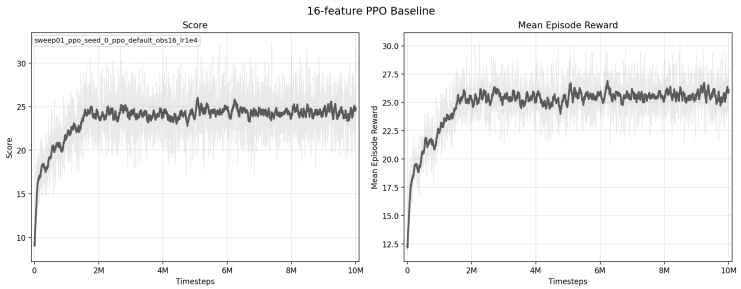
Action: $\alpha_t \in \mathbb{R}_{\geq 0}$, coefficient defining bids ($\text{bid}_i = \alpha_t \times \text{pValue}_i \times \text{target CPA}_i$)

Reward: $R = \text{conversion} \cdot \min\left\{1, \left(\frac{\text{CPA}}{\text{cost}}\right)^2\right\}$

Baseline Algorithm: PPO

Training and Evaluation:

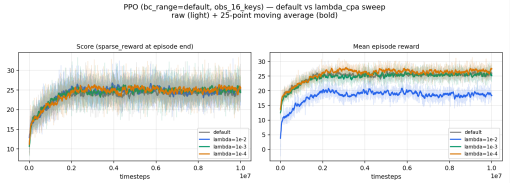
- Online training on 20 data periods.
- Evaluate on 1 independent period.



Extensions

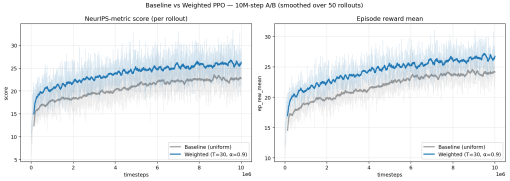
1. Reward shaping

Lagrangian Direct Penalty for CPA exceedance.



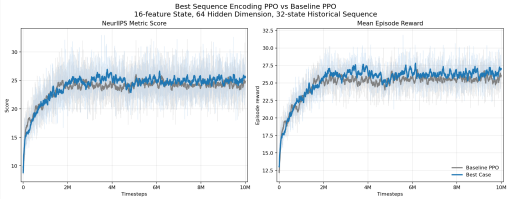
2. Replay reweighting

Reweight advertiser sampling with a softmax modification.



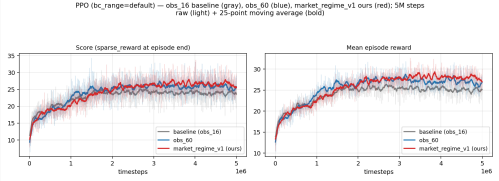
3. Sequence encoder (GRU)

Encode historical steps into a summarizing state via neural network.



4. Market-regime conditioning

Condition policy on inferred market competitiveness.



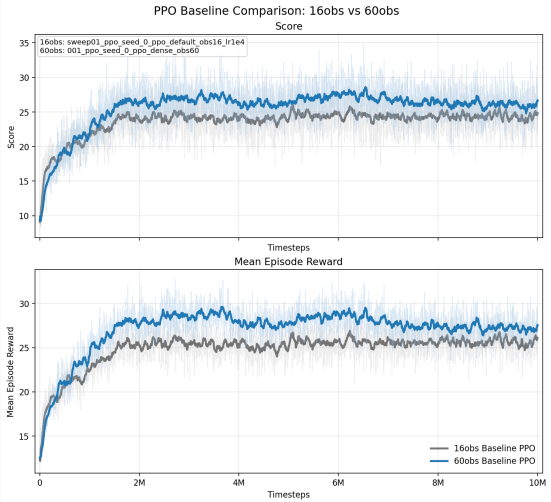
Extension: State Feature Augmentation

Motivation: AuctionNet winner uses 60-feature states.

60-feature states improves baseline PPO performance.

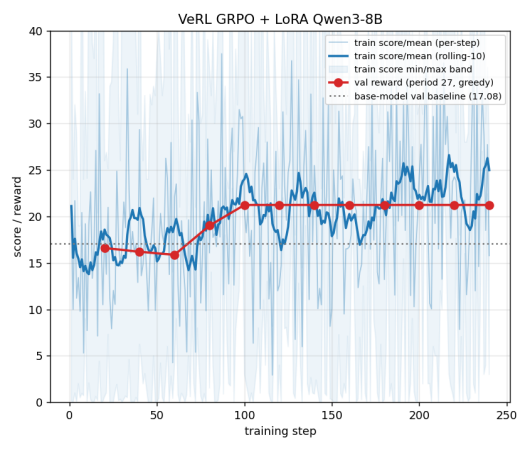
Why:

- Conditions policy on CPA target
- Adds historical bidding/pacing context
- Enables constraint-aware decisions



Extension: RL Training on LLM

- **Motivation:** Does reinforcement learning on a pretrained LLM give better sample efficiency than training a policy from scratch with PPO?
- **Setup:** GRPO + LoRA on Qwen3-8B
- **Findings:**
  - Early stage beats from-scratch PPO
  - plateaus at 21, below from-scratch PPO ceiling.
  - Trades peak performance for sample efficiency.



Evaluations & Next Step

Method	Score
PPO baseline	24.8
+ Replay reweight	25.8
+ GRU encoder	25.5
+ Market-aware	29.7
+ Reward shaping	25.6
+ 60-feature state	27.0
GRPO + LLM	21.2

- Several extensions show promising preliminary results.
- Due to limited computation and time, it remains unclear how much these extensions can improve performance.
- We will conduct a deeper hyperparameter search to further improve these results.